

Optimal Public Procurement Allocation: Balancing Sectoral Productivity and Competition ^{*}

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Abstract

This paper analyzes the optimal allocation of public procurement contracts in an oligopolistic economy. We develop a model in which a welfare-maximizing government selects its procurement strategy given an exogenous level of real public expenditure. Awarding procurement contracts alters market dynamics by affecting the markups of winning firms. While allocating contracts to the most productive firms seems optimal when considered in isolation, this strategy may not maximize welfare once sectoral and aggregate interactions are taken into account. The optimal allocation depends on the relative elasticities of substitution within and across sectors, as well as on the within sector dispersion of firm productivities. Under empirically plausible conditions, a more balanced distribution of contracts—rather than concentrating them among the most efficient firms—can enhance welfare. These findings highlight the role of procurement policy in shaping market structure, firm-level markups, and overall economic performance.

JEL: D43, D61, H57, L11, L50, E62

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1 Introduction

Public procurement is one of the largest and most consequential demand-side interventions in modern economies. In Europe alone, roughly one in seven euros flows through public tenders each year. In 2023, public buyers across the OECD–EU purchased goods and services amounting to approximately 14.8% of GDP, or about 2.5 trillion euros annually ([OECD, 2025](#)). At this scale, procurement policies do not merely determine which firms supply the public sector. By reallocating demand across suppliers, they reshape market shares, alter pricing incentives, and change markups in the private economy well beyond the public sector itself.

Over the past decade, procurement has also been explicitly repurposed as a policy lever. Award criteria increasingly go beyond the lowest invoice price to promote competition, SME participation, innovation, and the green transition. Strategic procurement plans at the EU level articulate these objectives, while national and regional authorities implement them through decentralized purchasing decisions. Yet this evolution raises a fundamental and unresolved question. Under modern procurement rules, the same public euro can either intensify competition or entrench market power. Which allocation of procurement awards actually lowers private prices, disciplines markups, and raises real incomes once markets adjust in general equilibrium?

This paper studies how a government should optimally design its procurement strategy—both within and across sectors—with particular emphasis on the within-sector allocation of awards when product markets are oligopolistic and markups are endogenous. We take seriously the institutional reality in which high-level procurement objectives are set centrally, while purchasing decisions are implemented by decentralized agencies that take these objectives as given. Against this backdrop, we ask how a welfare-maximizing government should deploy firm-level procurement weights once it internalizes how those weights reshape market structure through both public and private demand.

To address this question, we develop a tractable general-equilibrium model à la [Atkeson and Burstein \(2008\)](#) with two parallel demand systems: private demand from households and public demand from the government. On both sides, demand aggregates across sectors and across firms within sectors using standard nested constant-elasticity structures. Firms compete à la Cournot within sectors, implying endogenous markups that depend on market shares. The government acts as a Stackelberg leader. It chooses the procurement strategy—interpretable as eligibility rules, preferences, or scoring criteria—anticipating their general-equilibrium effects on prices, markups, and aggregate consumption. Procurement agencies then allocate spending to minimize expenditure subject to these rules, while households and firms take the resulting allocation as given.

The core mechanism is straightforward. In oligopolistic markets, a firm’s markup increases with the market share implied by the demand system. Tilting procurement toward a particular supplier

expands its total demand, raises its market share, and therefore increases its markup; rival firms experience the opposite effect. These changes feed directly into both the government’s procurement price index and the private consumption price index. The strength of this channel is governed by two elasticities: substitution across firms within a sector and substitution across sectors. Procurement reallocations therefore operate through the same channel as large private demand shocks, but with the crucial difference that they are policy-chosen.

Three main results emerge. First, introducing a welfare-maximizing government into an oligopolistic economy highlights that procurement reallocations have general-equilibrium effects on both public and private prices through endogenous markups. Government demand is not neutral: reallocating awards changes market power and thus affects the cost of living. Second, the optimal procurement rule equates the marginal impact of reallocating awards on the private consumption price index and on the public procurement price index. This condition makes explicit the central efficiency–competition trade-off: concentrating awards improves productive efficiency but weakens price discipline, while diversification fosters competition at the expense of some efficiency. Third, the welfare ranking of procurement regimes depends jointly on productivity dispersion and the gap between intra- and inter-sector elasticities. When productivity dispersion is modest, balanced allocations that preserve rivalry maximize welfare. When dispersion is high, the optimal policy becomes interior—tilting decisively, but not exclusively, toward the most productive suppliers. In general, value-for-money based solely on invoice prices is not the welfare-maximizing criterion.

These results speak directly to ongoing policy debates. Concentrating procurement can deliver lower public prices today but inflate markups and private prices tomorrow; spreading awards can discipline market power but raise public expenditure. By mapping procurement weights into sectoral price indices and real wages in general equilibrium, our framework clarifies when each approach dominates and yields testable predictions for environments in which governments pursue competition, efficiency, and strategic objectives simultaneously.

Our analysis relates to, and bridges, several strands of the literature. First, it builds on the macroeconomic theory of imperfect competition with endogenous markups, pioneered by [Atkeson and Burstein \(2008\)](#) and extended by [Edmond *et al.* \(2015, 2018\)](#) and [Burstein *et al.* \(2025\)](#). These contributions show how market shares and markups respond systematically to demand and productivity, with important implications for welfare. We extend this framework by endogenizing government demand and showing how procurement reallocations function as a policy-driven source of market share variation.

Second, the paper contributes to the growing literature on the macroeconomic effects of public procurement. Recent work highlights how government purchasing interacts with financial frictions, production networks, and fiscal transmission ([Barattieri *et al.*, 2021](#); [di Giovanni *et al.*, 2022](#); [Cox](#)

et al., 2023). In contrast to this literature, which typically treats procurement as an exogenous demand shifter, we model procurement allocation itself as a policy choice that reshapes market structure.

Third, our work connects to the empirical and policy-oriented literature on procurement design. Studies show that award rules and scoring criteria affect prices, participation, and firm performance (Decarolis, 2014; Bosio *et al.*, 2020, 2023; Moszoro and Spiller, 2022). Theoretical contributions emphasize procurement as an industrial policy tool (Celentani *et al.*, 2019; Veugelers, 2021). We provide a micro-founded general-equilibrium framework that rationalizes these insights by linking procurement rules directly to competition, markups, and welfare.

Overall, the paper reframes public procurement as a first-order competition policy instrument. By showing how procurement weights translate into market power and real incomes, it provides a disciplined way to evaluate procurement regimes that prioritize efficiency, competition, or strategic objectives in isolation—and to understand when such priorities are mutually reinforcing or fundamentally in tension.

2 Model with Endogenous Government Procurement

We extend the baseline multi-sector, oligopolistic competition model in Burstein, Carvalho and Grassi (2022) by incorporating a welfare-maximizing government that strategically allocates public procurement across sectors and firms. This introduces an additional layer of general equilibrium interaction, with procurement policy endogenously determined to maximize aggregate welfare.

2.1 The Role of the Government

A central innovation of our framework is the explicit and endogenous role of the government as a welfare-maximizing agent that shapes the equilibrium of the economy through its procurement strategy. This section details the government’s objective, the timing of its decisions, and how its choices influence market outcomes.

Government as a Welfare maximizer Unlike standard models in which government demand is either exogenous or passively determined, we model the economy as a Stackelberg game: The government acts as a welfare maximizer that (i) anticipating the behaviour of other agents, choose a procurement strategy (vector of B_{ki}) and (ii) its dependent agencies choose the procurement allocation to minimize expenditure, simultaneous to the household and firm’s problems. A real life example is the EU, that in 2017 established 6 strategic priorities for the following years ¹. Then, procurement agencies (such as national governments, ministries, universities, hospitals, etc) assume these preferences to award procurement contracts. These plans are set long term (approximately for the next 10 years) and include criterion such as: Green transition, social benefits or being a SME.

Procurement Allocation Problem. The government is endowed with a fixed real procurement budget G , which it allocates across sectors and firms through a two-level CES expenditure minimization problem, analogous to the household. The government’s allocation is characterized by two sets of CES weights: $\{B_k\}_{k=1}^N$ across sectors and $\{B_{ki}\}_{i=1}^{n_k}$ within each sector k . The firm-level demand shifters are not fixed, but are the policy tool that the government’s can use to affect competition. Government consumption and prices are aggregated following the standard CES procedure:

¹European Commission, Making Public Procurement Work in and for Europe (2017)

$$G_k = \left[\sum_i^{n_k} B_{ki}^{\frac{1}{\varepsilon}} G_{ki}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}, \quad G = \left[\sum_k^N B_k^{\frac{1}{\sigma}} G_k^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

Given these, the government's procurement demand at the firm level is (see [Appendix B](#)):

$$G_{ki} = B_k B_{ki} P_{ki}^{-\epsilon} (P_k^G)^{\epsilon-\sigma} (P^G)^\sigma G,$$

where P^G is the government price index and (P_k^G) is the government sectoral price index.

$$P_k^G = \left[\sum_i^{n_k} B_{ki} (P_{ki})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \quad P^G = \left[\sum_k^N B_k (P_k^G)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Objective and Constraints. The government's objective is to maximize the equilibrium value of aggregate private consumption, C , by choosing the vector $\{B_{ki}\}$, subject to:

- The Household's budget Constraint: $P^C C = W - T$, where T is tax revenue.
- Its own balanced budget constraint: $P^G G = T$
- The entire system of general equilibrium constraints, which includes the optimizing behavior of households and firms, market clearing, and the feedback effects of procurement allocations on equilibrium prices and quantities.

Timing Crucially, the government chooses its procurement allocation $\{B_{ki}\}$ *anticipating* the endogenous responses of both firms and households:

1. The government sets $\{B_{ki}\}$ to maximize aggregate consumption, taking into account the induced general equilibrium.
2. Given these procurement weights, both the household and procurement agencies solve their respective expenditure minimization problems, generating demand schedules for each firm. Simultaneously, firms set prices à la Cournot, determining endogenous markups and quantities.

Summary. In summary, the government in our model is an active and forward-looking policy maker whose procurement decisions are central to the determination of equilibrium outcomes. By moving first and internalizing the responses of households and firms, the government is able to exploit procurement as a powerful policy instrument for enhancing welfare in oligopolistic markets.

2.2 Household Demand

In the private sector, Household takes consumption decisions at firm level by minimizing expenditure. As with the public sector, goods (and therefore prices) are aggregated into sectors and then into the economy following a double CES aggregation system:

$$C_k = \left[\sum_i^{n_k} A_{ki}^{\frac{1}{\epsilon}} C_{ki}^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \quad C = \left[\sum_k^N A_k^{\frac{1}{\sigma}} C_k^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

The household chooses consumption $\{C_{ki}\}$ to minimize expenditure for a given utility level:

$$\begin{aligned} \min_{\{C_{ki}\}} & \sum_{k=1}^N \sum_{i=1}^{n_k} P_{ki} C_{ki} \\ \text{s.t.} & \quad C \geq \bar{C} \end{aligned}$$

where C is a two-level CES aggregator (see [Appendix A](#) for details). The corresponding demand function is:

$$C_{ki} = A_k A_{ki} P_{ki}^{-\epsilon} (P_k^C)^{\epsilon-\sigma} (P^C)^{\sigma} C,$$

where P_C is the aggregate consumption price index and P_k is the sectoral price index.

$$P_k^C = \left[\sum_i^{n_k} A_{ki} (P_{ki})^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}, \quad P^C = \left[\sum_k^N A_k (P_k^C)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

2.3 Firm Problem and Endogenous Markups

Each firm (k, i) chooses price P_{ki} to maximize profits:

$$\begin{aligned} \max_{P_{ki}} & \{ (P_{ki} - w/Z_{ki}) \cdot Y_{ki} \} \\ \text{s.t.} & \quad Y_{ki} = C_{ki} + G_{ki} \end{aligned}$$

Where C_{ki} and G_{ki} are the public and private demands as computed in the previous minimum expenditure problems, w is the wage and Z_{ki} is firm productivity.

The optimal markup as shown in (Appendix C) :

$$\mu_{ki} = (1 - \gamma_{ki}^{-1})^{-1} = \frac{\varepsilon}{\varepsilon - 1} \left[1 - \frac{\frac{\varepsilon}{\sigma} - 1}{\varepsilon - 1} \frac{S_{ki}C_{ki} + S_{ki}^G G_{ki}}{Y_{ki}} \right]^{-1}$$

where γ_{ki} is the elasticity of the firm's total demand with respect to its own price, aggregating both private and government demand.

This extended framework highlights the general equilibrium effects of endogenous government procurement. By choosing procurement shares strategically, the government can influence equilibrium market shares, markups, and ultimately aggregate consumption. The interplay between public and private demand, and its effect on market power, emerges endogenously from the equilibrium system.

All formal derivations and technical proofs are provided in the Appendix.

2.4 Government Procurement Strategy

This Stackelberg structure means that the government's procurement strategy internalizes all general equilibrium repercussions, including effects on market power, sectoral output, and price indices.

The government aims to maximize aggregate private consumption C by choosing firm-level procurement weights $\{B_{ki}\}$, subject to satisfying the real public consumption needs in the economy. G is an exogenously set target that has to be consumed by the Government. In this paper, we do not take a stance on how does the Government calculate the real needs of the economy.

$$\max_{\{B_{ki}\}} C = \frac{\omega - P_G G}{P_C} \quad \text{s.t.} \quad \begin{cases} G = \bar{G}, \\ P^G = \left[\sum_k^N B_k (P_k^G)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, & P_k^G = \left[\sum_i^{n_k} B_{ki} (P_{ki})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \\ P^C = \left[\sum_k^N A_k (P_k^C)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, & P_k^C = \left[\sum_i^{n_k} A_{ki} (P_{ki})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}} \end{cases}$$

First-Order Condition and Characterization. The government's optimization yields a system of first-order conditions characterizing the optimal procurement allocation. For each B_{ki} , the key necessary condition (see Appendix for derivation) relates the marginal effect of procurement on the government and household price indices:

$$\frac{P_G G}{P_C C} = -\frac{\Phi_{ki}^C}{\Phi_{ki}^G} = -\frac{\sum_h \sum_j S_h^C S_{hj}^C \phi_{hj,ki}}{\sum_h \sum_j S_h^G S_{hj}^G \phi_{hj,ki} + S_k^G S_{ki}^G}$$

where $\Phi_{ki}^C = \frac{\partial P^C}{\partial B_{ki}} \frac{B_{ki}}{P^C}$ and $\Phi_{ki}^G = \frac{\partial P^G}{\partial B_{ki}} \frac{B_{ki}}{P^G}$ are the elasticities of the consumption and government price indices with respect to B_{ki} , respectively. Furthermore, $\phi_{hj,ki} = \frac{\partial P_{hj}}{\partial B_{ki}} \frac{B_{ki}}{P_{hj}}$ is the firm level price elasticity with respect to B_{ki} . Proofs can be seen in [Appendix D](#) and [Appendix F](#). This condition states that the government allocates procurement so that the (negative) marginal impact on the cost of private consumption, relative to its own procurement cost, is equated across all procurement options, taking into account both direct and general equilibrium effects. Notice its own procurement cost is directly related to the tax collection. If the Government does not take into account aggregate efficiency, the price for the contract will be higher, and more taxes will have to be collected from the consumer as a consequence, which will negatively affect aggregate consumption. However, awarding contracts only to the most productive firm will increase aggregate markups, affecting the purchasing power of the consumer.

Economic Interpretation. By endogenously allocating its procurement, the government can tilt equilibrium allocations to sectors and firms in a way that reduces market power (by favoring competitive or more elastic sectors), enhances overall consumption, and improves welfare relative to passive or exogenous procurement. The government thus acts as an active industrial policy maker, using procurement not only as a means of acquiring goods, but as a lever to influence the structure and efficiency of the entire economy.

Two primitives govern the trade-off. First, the *space for strategic pricing*, captured by the gap between the within-sector elasticity ε and the across-sector elasticity σ . When $\varepsilon = \sigma$ (monopolistic-competition benchmark), markups are effectively constant and procurement only selects on productivity; as ε rises above σ , procurement-induced share shifts feed into markups, strengthening the competition channel. Second, *productivity dispersion* within the sector determines the strength of selection: larger gaps in productivity magnify the invoice-price gains from tilting awards toward the leader.

3 Results: A simple case

In this section, we present the numerical solution of the model and discuss various quantitative results. To clearly understand the trade-off of our model, we show the results for a two sector and two firms economy. This allows us to solve numerically the problem of the firm, the consumer and the government and delivers clear results on the effect of procurement weights on market structure, prices and welfare. Then, we discuss four different scenarios, according to the space for strategic pricing (monopolistic vs oligopolistic competition) and the productivity dispersion.

When the government sets the procurement strategy, she needs to take into account the effects on all the variables of the market. In particular, supporting the most productive firm has two immediate effects: (i) increase of its markup, which increases private market price given it is the dominating firm and (ii) reshuffles the public sector market shares, decreasing procurement aggregate price (and therefore taxes) due to higher sectoral productivity. On the other hand, supporting the less productive firm, can equalize markups and improve competition in the private market, but at the same time reduces overall productivity. In this scenario, it is important to understand not only the dispersion in productivities but also the substitutability between sectors and within sector.

3.1 Mechanism in action

In this subsection we proceed to discuss the equilibrium dynamics as the government changes the procurement strategy in the baseline scenario. First, we comment the direct effect on markups and market shares, followed by how that affects sectoral and aggregate variables. Then, we conclude with the optimal procurement strategy in this baseline scenario. We consider two sectors with two firms each. For simplicity we assume symmetric sectoral weights from both the consumers' and the Government's perspective ($A_{ki} = 1$ and $A_k = B_k = 1$) and normalise $\omega = 1$ and $G = 1$. Firms in sector 2 are symmetric, $[Z_{2i}] = 1 \forall i$. In the sector 1, firm 2 is more productive, with $\{Z_{11}, Z_{12}\} = \{1, 2\}$. We choose $\sigma = 1.8$ and $\varepsilon = 5$ following the calibration in [Burstein *et al.* \(2025\)](#).

Under this calibration, we conduct two exercises to understand the mechanisms of our model. First, we study the subgame played by the firms and the consumers for different procurement weights. We vary sector one procurement weights to see its effects on the different parameters. Second, we study the solution of the government's problem. We find numerically the optimal procurement weights for every firm that maximize the aggregate consumption and analyse its dependence on parameters.

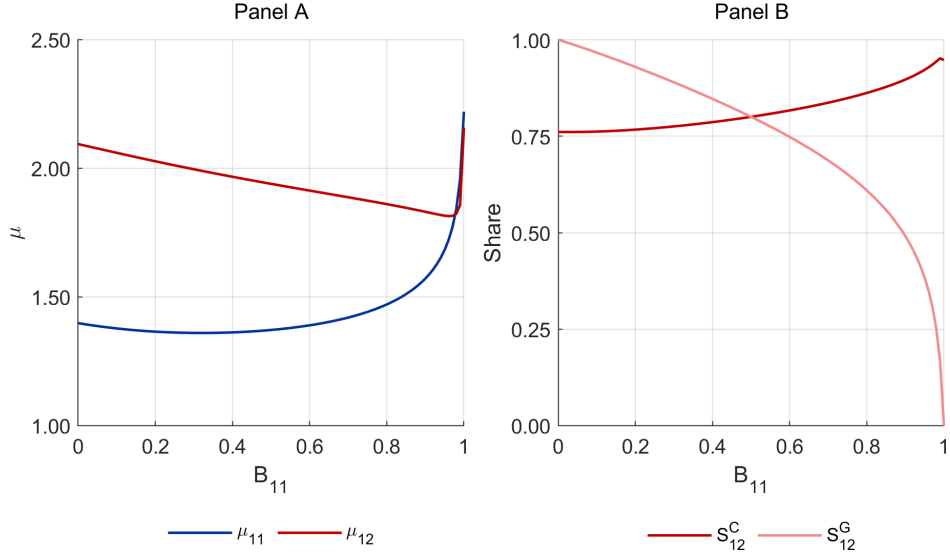


Figure I – Sector 1 markups, market shares, and production ratios as B_{11} varies.

Figure I documents the reaction of markups and market shares, which are at the core of the aggregate effects discussed later. If the government assigns no procurement weight to firm one ($B_{11} = 0$) the government consumption for the sector is fully concentrated, so $s_{12}^G = 1$. As firm 2 is also the most productive firm, it is also able to capture a big part of the private market ($s_{12}^C \approx 0.75$) as we can see in Panel B. This is translated to markups, which are close to its maximum as we can see in the top-right graph. Firm 12 only has a higher markup when completely excluded from the public market, given it is allowed to fully exploit its market power advantage in the private market without needing to be competitive in the public one.

As B_{11} increases, public orders reallocate to the laggard. Two adjustments follow. First, the composition of sales tilts: C_{11}/Y_{11} falls and C_{12}/Y_{12} rises as 1 serves more of the public market while 2 pivots toward private customers. Second, private shares move in the opposite direction over most of the range: the non-recipient gains private ground ($s_{12}^C \uparrow$ when B_{11} is low to moderate) while the recipient cedes it. These share movements feed into markups: μ_{12} declines over most of the path as its total share shrinks, whereas μ_{11} eventually rises as its total (public+private) share expands. Near the corner $B_{11} \rightarrow 1$, 1 nearly monopolises procurement ($s_{11}^G \rightarrow 1$) and its markup spikes, while 2's markup ticks up slightly as it retreats into a small private niche. Figure I thus illustrates the core mechanism: procurement reallocate *both* public and private shares, which translates those changes into markups through oligopolistic competition.

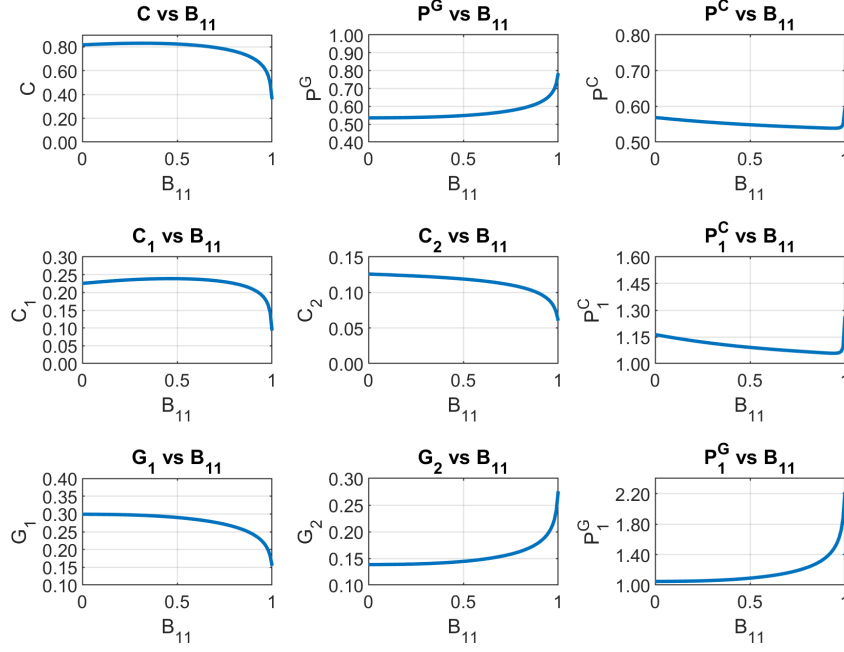


Figure II – Sectoral and aggregate variables evolution as B_{11} varies.

Figure II maps these firm-level responses into sectoral and aggregate indices. Starting from low B_{11} , the sector-1 private index P_1^C falls as selection on the high- Z supplier dominates the initial change in μ_{11} ; the aggregate private index P^C declines and real consumption $C = (\omega - P^G G)/P^C$ rises. As procurement becomes highly concentrated on 1, the feedback from shares to markups strengthens: P_1^C turns up as the aggregator loads increasingly on a high-markup firm, and earlier gains unwind. The profile is a pronounced “V” in P^C (and an inverted “V” in C), delivering an interior consumption-maximising allocation around $B_{11} \approx 0.7$ in the baseline. Two features guide the interpretation. First, sector 2 is symmetric and its private index P_2^C is not affected by the procurement weight allocation in sector 1, so the movement in P^C is pinned down by sector 1; correspondingly, most of the non-linearity in C comes from C_1 . Second, the fiscal side works in the opposite direction: because $T = P^G G$, tilting procurement toward a given supplier raises the public index P^G via the direct weight effect even as rivals’ prices fall; this is the sign pattern $(\Phi_{ki}^C < 0, \Phi_{ki}^G > 0)$ embedded in the government FOC.

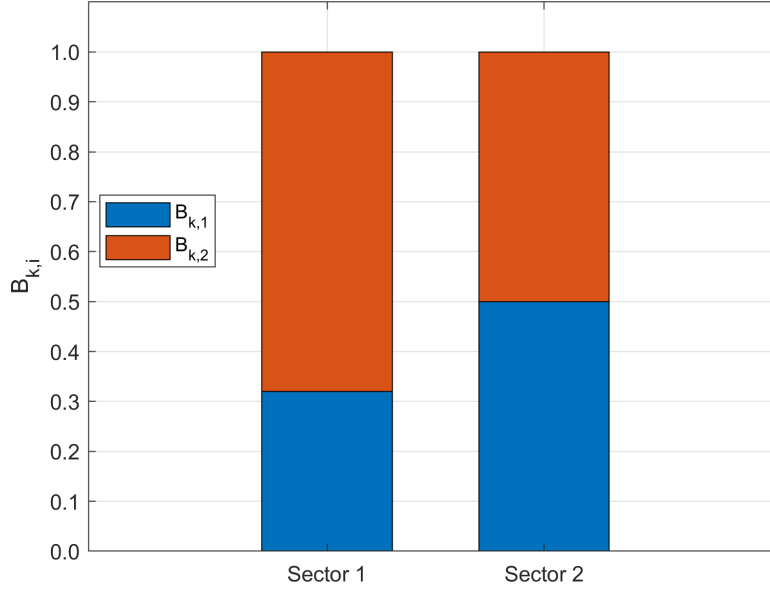


Figure III – Optimal B_{ki} allocation

Figure III summarises the welfare-maximising allocation. With symmetry in sector 2, equal splitting is efficient: any deviation expands a firm’s total share, raises its desired markup, and lifts the sectoral price index. In sector 1, the optimal mix assigns the majority of the procurement budget to the productive leader (2) but retains a sizeable share (about 30%) for the laggard (1), indicating that the least productive firm’s presence is valuable to discipline markups faced by both private and public buyers.

In the baseline scenario, under oligopolistic competition and high productivity dispersion, we have shown that it is optimal to distribute procurement between the two firms. Fully prioritizing the most productive firm leads to high markups that translate to high private market prices. On the other hand, including only the least productive firm translates to higher public market prices and therefore increasing taxes (and lower disposable income) for the consumers. As we have shown, the productivity channel is more powerful, implying that it is optimal to give a lower (but non-zero) weight to the least productive firm.

3.2 Effect of Market structure: Strategic pricing vs productivity dispersion

We now compare the four scenarios implied by the interaction of our two primitives: (i) *productivity dispersion* within a sector (low vs. high) and (ii) the *space for strategic pricing*, captured by the gap between the within- and across-sector elasticities ($\varepsilon - \sigma$). We show that under monopolistic competition ($\varepsilon \approx \sigma$), markups are independent of any decision variable in the model: when

productivities are close the welfare ranking is nearly flat and balanced splits are innocuous, while under *high dispersion* the optimal rule is to concentrate on the most productive firm—i.e., the lowest-price rule (pure selection with no competition downside). In the oligopolistic competition scenario, $\varepsilon \gg \sigma$), share shifts move markups: under *low dispersion* preserving rivalry dominates—split awards to keep both firms in play and discipline prices; under *high dispersion* the optimum is interior—tilt decisively toward the leader for selection gains but keep a positive share for the follower to curb the leader’s markup, thus avoiding corner allocations. In what follows we first describe the mechanism and provide an overview of equilibrium behaviour; we then compare the four scenarios and show why the same intervention can have sharply different effects across them.

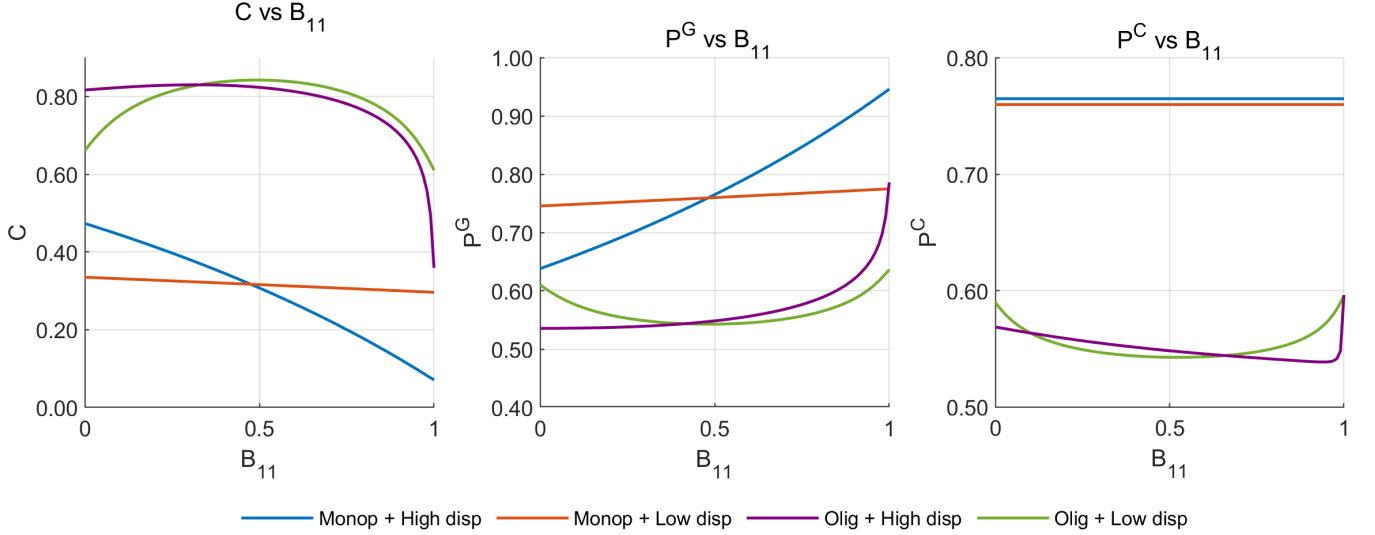


Figure IV – Aggregate variables evolution by case

Figure IV shows how the aggregate variables evolve with procurement strategy. Under *monopolistic competition* ($\varepsilon = \sigma$), private prices are flat and at a higher level than under oligopoly, and the public price index moves little. With markups insensitive to shares, procurement cannot discipline market power, leaving scant reason to deviate from a minimum-price rule that concentrates awards on the most productive supplier. In this environment, including the laggard when dispersion is high mainly raises the public price index and—via the budget identity—reduces consumption. By contrast, *oligopolistic competition* ($\varepsilon > \sigma$) introduces the central non-linear trade-off. With high dispersion, the lowest private price occurs near an extreme tilt toward the leader (about 97% of the procurement weight to firm 2). Exactly there, however, the public price index peaks, implying higher taxes that erode private-price gains; the consumption panel shows the net effect. Across cases, the highest aggregate consumption arises under oligopoly with *low* dispersion and an *equal split* of procurement weight. With *high* dispersion, the optimum remains interior: allocating a meaningful yet limited share to the laggard delivers higher consumption than corner allocations by balancing selection gains against markup discipline.

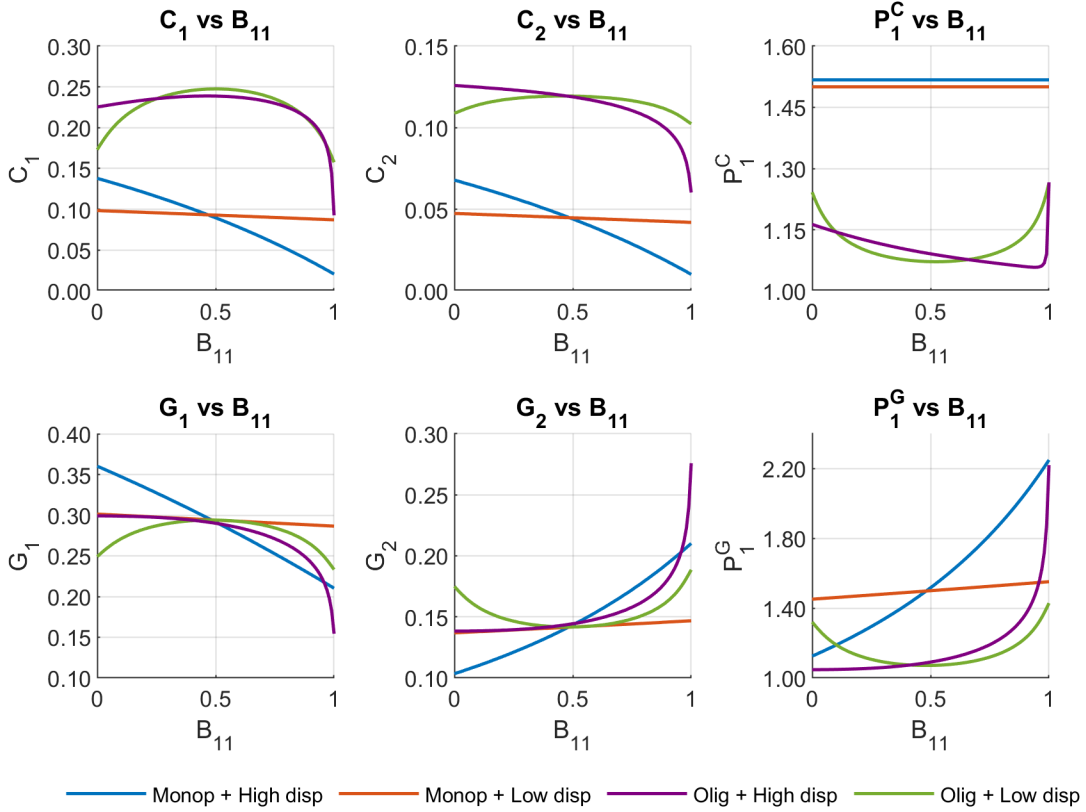


Figure V – Sectoral variables evolution by case

Figure V shows the sectoral origin of the aggregate changes discussed before. In sector 1—the sector whose awards are reallocated—the private index P_1^C is essentially flat under monopolistic competition (blue, orange) but non-monotone under oligopoly (purple, green): it falls initially as selection dominates, then rises when concentration feeds back into markups. The public index P_1^G moves in the opposite direction because the sectoral government aggregator loads directly on the award weights. These movements generate the hump in C_1 for moderate tilts and the subsequent decline near the corner. Sector 2, untouched by the B_{11} change, adjusts through general equilibrium: C_2 gently drifts as income and resources shift with P^C and P^G , and G_2 rises with B_{11} to satisfy the fixed G , with curvature strongest under oligopoly and high dispersion.

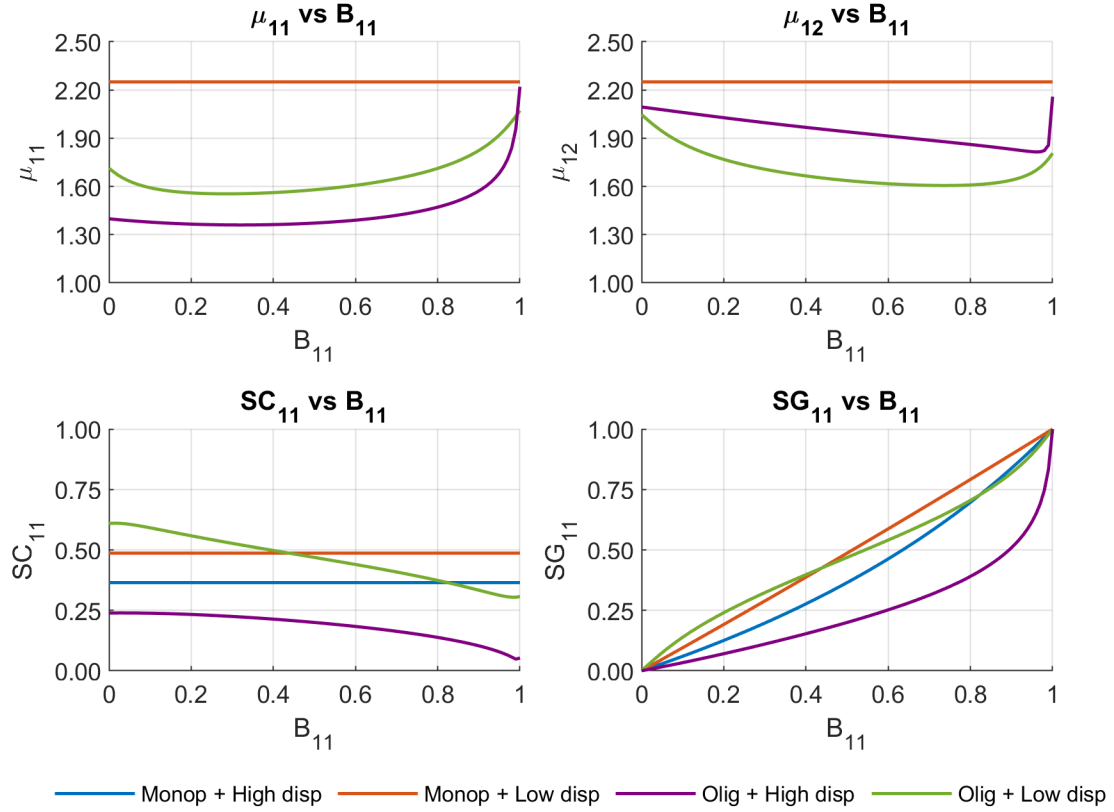


Figure VI – Markups and market shares evolution by case

Firm-level responses in Figure VI align with the sectoral curves. Under monopolistic competition (blue, orange), μ_{11} and μ_{12} are essentially constant and private shares (SC_{11}, SC_{12}) barely move—procurement cannot discipline market power. Under oligopoly (purple, green), markups inherit the curvature of shares. As B_{11} rises, public shares move one-for-one by construction, while private shares move in the *opposite* direction over most of the range, reflecting rivalry in the private market. With *high dispersion* (purple) these reallocations are steep, producing pronounced U- and inverted-U shapes in μ_{11} and μ_{12} and a spike for μ_{11} near $B_{11} \rightarrow 1$. With *low dispersion* (green) the same forces operate with milder slopes, yielding smooth interior optima in prices and consumption.

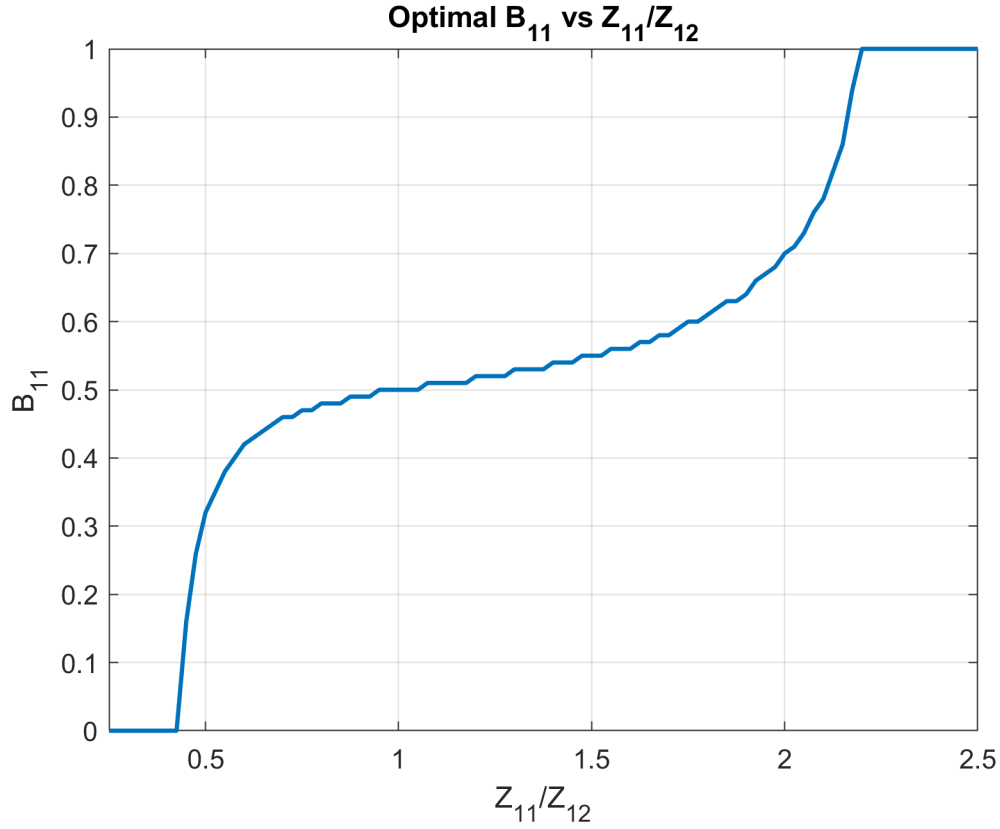


Figure VII – Optimal procurement weight allocation by case

Figure VII plots the welfare-maximizing procurement weight assigned to firm 1 as a function of the relative productivity ratio Z_{11}/Z_{12} . When this ratio is below one, firm 1 is the less productive supplier; when it exceeds one, it becomes the more productive one. The figure shows that for sufficiently large productivity gaps, the optimal allocation is a corner solution in which all procurement is directed to the more productive firm, as splitting contracts would generate efficiency losses that dominate any competitive benefits. Only when the two firms are nearly symmetric in productivity does the planner choose a balanced allocation with $B_{11} \approx 0.5$, consistent with diversification being most valuable when firms have comparable efficiency. In the intermediate region, where productivity differences are moderate, B_{11} increases gradually with Z_{11}/Z_{12} : once firm 1 becomes productive enough to merit participation, its share rises smoothly until a sharp increase emerges as it approaches the point where it is sufficiently more productive than its rival. Overall, the pattern highlights how procurement weights respond smoothly to small productivity differences but shift rapidly when one firm becomes decisively more efficient.

4 Conclusion

This paper develops a tractable general-equilibrium model in which a welfare-maximizing government uses public procurement as a strategic instrument to shape market outcomes in oligopolistic industries. The model formalizes a central policy tension: allocating awards exclusively to the most productive firms reduces the public sector’s procurement prices but can simultaneously erode competition, raise markups, and increase the private cost of living. Conversely, spreading awards more evenly restrains market power but may raise procurement expenditure. Our framework provides an analytical lens through which to understand and quantify this *value-for-money versus competition* trade-off and identifies the conditions—expressed in elasticities and expenditure shares—under which optimal procurement is concentrated or balanced.

At the core of the analysis lies a simple yet powerful mechanism. Because firms compete à la Cournot, their desired markups rise with total demand shares. When the government tilts procurement toward one supplier, it expands that firm’s share and markup, while compressing those of its rivals. These changes propagate to both the government’s procurement price index and the private consumption price index. The first-order condition derived from the government’s optimization problem captures this dual effect: at the optimum, the marginal (negative) impact of procurement on the private price index equals its (positive) impact on the public price index, weighted by expenditure shares. This elasticity-based representation not only clarifies the logic of the optimal allocation but also renders it empirically measurable.

Our quantitative analysis illustrates how these forces interact. In a stylized two-sector, two-firm economy, we show that procurement reallocations have non-linear effects on prices and welfare. Under monopolistic competition—where within- and across-sector elasticities coincide—markups are constant, so procurement simply selects on productivity. In this case, concentrating awards on the most productive firm minimizes procurement prices and maximizes welfare. Under oligopolistic competition, by contrast, markups respond endogenously to demand shares, creating a strategic pricing channel. When productivity dispersion is low, balanced awards that preserve rivalry yield the greatest welfare gains. When dispersion is high, the optimal allocation becomes interior: the government should tilt decisively toward the productive leader but maintain a positive share for the follower to contain market power. Thus, procurement performs best as a *graded policy* rather than a binary rule.

The broader implication is that procurement policy can operate as an instrument of competition policy. By allocating contracts across firms with an eye to their effects on market structure, governments can raise real incomes without necessarily increasing public spending. In practical terms, the model’s predictions translate into actionable design principles: scoring rules that value

both efficiency and rivalry, caps on within-lot volumes, and the deliberate use of multiple suppliers can help implement near-optimal allocations. When elasticities and market shares are observable, the first-order condition offers a diagnostic criterion: if the marginal reduction in the private cost of living is smaller (in absolute value) than the marginal rise in the public procurement index, the current allocation is too concentrated.

The framework also yields clear empirical predictions. Within sectors, increases in a firm's procurement share should raise its markup and lower its rivals' markups; sectoral private price indices should initially fall and then rise as procurement becomes concentrated. Across sectors, welfare improvements from diversification are strongest when within-sector substitution is limited and productivity gaps are moderate. These patterns can be tested in micro data using award reallocations, scoring thresholds, or procurement reforms as quasi-experiments.

Future work should extend the model's scope along dynamic and structural dimensions. Allowing for entry, capacity investment, or learning-by-doing would enrich the set of channels through which procurement affects long-run competition and productivity. Moreover, integrating richer contract structures—framework agreements, quality-adjusted scoring, or green and social criteria—would further align the theoretical framework with the complexity of real procurement strategies. Despite these simplifications, the current setup delivers a clear and actionable insight: *effective procurement policy must balance selection on productivity with the preservation of competition*. Governments can thus use procurement not merely as a spending instrument, but as a lever to foster more efficient and competitive markets.

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Appendix

A Household Problem

The representative Household allocates her consumption across different sectors to minimize total expenditure while ensuring that aggregate consumption meets a required utility level.² We divide the problem in two parts: within and across sectoral decisions.

A.1 Inter-sectoral Decision

The household solves the following problem:

$$\begin{aligned} \min_{\{C_k\}_{k=1}^N} \quad & \sum_k P_k C_k \\ \text{s.t.} \quad & \left[\sum_{j=1}^N A_j^{\frac{1}{\sigma}} C_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \geq C \end{aligned}$$

The Lagrangian associated with this problem is:

$$\mathcal{L}(C_k, \mu) = \sum_k P_k C_k - \mu \left[\left[\sum_{j=1}^N A_j^{\frac{1}{\sigma}} C_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} - C \right]$$

The first-order condition with respect to C_k is:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial C_k} = 0 & \Rightarrow P_k - \mu \left[\sum_{j=1}^N A_j^{\frac{1}{\sigma}} C_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} A_k^{\frac{1}{\sigma}} C_k^{\frac{-1}{\sigma}} = 0 \Rightarrow \\ & \Rightarrow P_k = \mu C^{\frac{1}{\sigma}} A_k^{\frac{1}{\sigma}} C_k^{\frac{-1}{\sigma}} \end{aligned}$$

Solving for C_k , we obtain:

²As $f_0 = 0$ this is equivalent to a certain consumption level

$$C_k = \mu^\sigma P_k^{-\sigma} A_k C$$

Since the CES aggregator satisfies homogeneity of degree 1, we have:

$$\sum_k P_k C_k = PC$$

Substituting C_k :

$$\sum_k P_k \left(\mu^\sigma P_k^{-\sigma} A_k C \right) = PC \Rightarrow \mu^\sigma C \sum_k A_k P_k^{1-\sigma} = PC \Rightarrow$$

$$\Rightarrow \mu^\sigma \sum_k A_k P_k^{1-\sigma} = P \Rightarrow \mu = P$$

Thus, the intersectoral inverse demand function follows as:

$$C_k = P^\sigma P_k^{-\sigma} A_k C$$

A.2 Within-Sector Decision

Given C_k , the household further allocates consumption among firms within each sector by solving:

$$\begin{aligned} \min_{\{C_{kj}\}_{j=1}^{N_k}} & \sum_{j=1}^{N_k} P_{kj} C_{kj} \\ \text{s.t.} & \left[\sum_{j=1}^{N_k} A_{kj}^{\frac{1}{\varepsilon}} C_{kj}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} \geq C_k \end{aligned}$$

The Lagrangian is:

$$\mathcal{L}(P_{kj}, \lambda) = P_{kj} C_{kj} - \lambda \left(\left[\sum_{j=1}^{N_k} A_{kj}^{\frac{1}{\varepsilon}} C_{kj}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} - C_k \right)$$

Taking the derivative with respect to C_{kj} :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial C_{kj}} = 0 &\Rightarrow P_{kj} - \lambda \left[\sum_{j=1}^{N_k} A_{kj}^{\frac{1}{\varepsilon}} C_{kj}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{1}{\varepsilon-1}} A_{kj}^{\frac{1}{\varepsilon}} C_{kj}^{-\frac{1}{\varepsilon}} = 0 \\ &\Rightarrow P_{kj} = \lambda C_k^{\frac{1}{\sigma}} A_{kj}^{\frac{1}{\sigma}} C_{kj}^{-\frac{1}{\sigma}}\end{aligned}$$

Solving for C_{kj} :

$$C_{kj} = \lambda^{\varepsilon} C_k A_{kj} P_{kj}^{-\varepsilon}$$

Since the CES aggregator satisfies homogeneity of degree 1, we have:

$$\sum_k P_{kj} k C_{kj} = P_k C_k$$

and similar to the intersectoral problem, we get $\lambda = P_k$, and substitute:

$$C_{kj} = P_k^{\varepsilon} C_k A_{kj} P_{kj}^{-\varepsilon}$$

A.3 Private sector demand function

Now, combining the intersectoral and intrasectoral demands, we derive the final inverse demand function from the Household:

$$C_{kj} = A_k A_{kj} P^{\sigma} P_{kj}^{-\varepsilon} P_k^{\varepsilon-\sigma} C$$

B Government Problem

The Government allocates its expenditure across sectors similarly to the Household.

B.1 Inter-sectoral Decision

The Government solves:

$$\begin{aligned} \min_{\{G_k\}_{k=1}^N} \quad & \sum_k P_{k,G} G_k \\ \text{s.t.} \quad & \left[\sum_{j=1}^N B_j^{\frac{1}{\sigma}} G_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} = G \end{aligned}$$

Following a similar derivation to the Household Problem, the Government's intrasectoral demand is:

$$G_k = P_G^\sigma P_{k,G}^{-\sigma} B_k G$$

B.2 Within-Sector Decision

Within each sector, the Government further allocates spending among firms, solving:

$$\begin{aligned} \min_{\{G_{kj}\}_{j=1}^{N_k}} \quad & \sum_{j=1}^{N_k} P_{kj} G_{kj} \\ \text{s.t.} \quad & \left[\sum_{j=1}^{N_k} B_{kj}^{\frac{1}{\varepsilon}} G_{kj}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} = G_k \end{aligned}$$

Solving as before, the intra-sectoral Government demand is:

$$G_{kj} = P_{k,G}^\varepsilon G_k B_{kj} P_{kj}^{-\varepsilon}$$

B.3 Public sector demand function

Now, combining the intersectoral and intrasectoral demands, we derive the final inverse demand function for the public sector:

$$G_{kj} = B_k B_{kj} P_G^\sigma P_{kj}^{-\varepsilon} P_{k,G}^{\varepsilon-\sigma} G$$

C Firm problem

The firm seeks to maximize its profit, which is given by the price-cost margin multiplied by the output level:

$$\max_{Y_{ki}} (P_{ki} - \tilde{W}_{ki}) Y_{ki}$$

To find the optimal output level, we differentiate the profit function with respect to Y_{ki} and set the derivative equal to zero:

$$(P_{ki} - \tilde{W}_{ki}) + \frac{\partial P_{ki}}{\partial Y_{ki}} Y_{ki} = 0 \Rightarrow 1 + \frac{\partial P_{ki}}{\partial Y_{ki}} \frac{Y_{ki}}{P_{ki}} = \frac{\tilde{W}_{ki}}{P_{ki}} \Rightarrow P_{ki} = \tilde{W}_{ki} (1 - \gamma_{ki}^{-1})^{-1}$$

where $\gamma_{ki} = -\frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}}$ is the the price elasticity of demand.

From the constraint, the total demand elasticity can be decomposed into private and Government consumption components:

$$\frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}} = \frac{\partial C_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}} + \frac{\partial G_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}}$$

where:

$$C_{ki} = P^\sigma C A_{ki} A_k P_{ki}^{-\varepsilon} P_k^{\varepsilon-\sigma}$$

$$G_{ki} = P_G^\sigma G B_{ki} B_k P_{ki}^{-\varepsilon} P_{k,G}^{\varepsilon-\sigma}$$

Differentiating C_{ki} with respect to P_{ki} and simplifying:

$$\begin{aligned} \frac{\partial C_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}} &= \frac{P_{ki}}{Y_{ki}} P^\sigma C A_{ki} A_k \left[-\varepsilon P_{ki}^{-\varepsilon-1} P_k^{\varepsilon-\sigma} + (\varepsilon - \sigma) P_{ki}^{-\varepsilon} P_k^{\varepsilon-\sigma-1} \frac{\partial P_k}{\partial P_{ki}} \right] = \\ &= \frac{P_{ki}}{Y_{ki}} P^\sigma C A_{ki} A_k P_{ki}^{-\varepsilon} P_k^{\varepsilon-\sigma} \left[-\varepsilon P_{ki}^{-1} + (\varepsilon - \sigma) \frac{1}{P_k} \frac{\partial P_k}{\partial P_{ki}} \right] = \frac{C_{ki}}{Y_{ki}} \left[-\varepsilon + (\varepsilon - \sigma) \frac{P_{ki}}{P_k} \frac{\partial P_k}{\partial P_{ki}} \right] \end{aligned}$$

Similarly, for Government demand:

$$\frac{\partial G_{ki}}{\partial P_{ki}} \frac{P_{ki}}{G_{ki}} = \frac{G_{ki}}{Y_{ki}} \left[-\varepsilon + (\varepsilon - \sigma) \frac{P_{ki}}{P_{k,G}} \frac{\partial P_{k,G}}{\partial P_{ki}} \right]$$

Combining both components:

$$\begin{aligned}\gamma_{ki} &= \frac{C_{ki}}{Y_{ki}} \left[-\varepsilon + (\varepsilon - \sigma) \frac{P_{ki}}{P_k} \frac{\partial P_k}{\partial P_{ki}} \right] + \frac{G_{ki}}{Y_{ki}} \left[-\varepsilon + (\varepsilon - \sigma) \frac{P_{ki}}{P_{k,G}} \frac{\partial P_{k,G}}{\partial P_{ki}} \right] \\ &= \varepsilon - \frac{\varepsilon - \sigma}{Y_{ki}} \left[C_{ki} \frac{P_{ki}}{P_k} \frac{\partial P_k}{\partial P_{ki}} + G_{ki} \frac{P_{ki}}{P_{k,G}} \frac{\partial P_{k,G}}{\partial P_{ki}} \right]\end{aligned}$$

Now, notice the elasticities can be decomposed as follows:

$$\frac{\partial P_k}{\partial P_{ki}} \frac{P_{ki}}{P_k} = \frac{\partial P_k}{\partial C_k} \frac{\partial C_k}{\partial Y_{ki}} \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{C_k}{P_k} \frac{P_k}{P_{ki}} \frac{P_{ki}}{Y_{ki}} = \frac{\partial P_k}{\partial C_k} \frac{C_k}{P_k} \frac{\partial C_k}{\partial Y_{ki}} \frac{Y_{ki}}{C_k} \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}}$$

where **(Proofs pending)**:

$$\frac{\partial P_k}{\partial C_k} \frac{C_k}{P_k} = -\frac{1}{\sigma} \quad \frac{\partial C_k}{\partial Y_{ki}} \frac{Y_{ki}}{C_k} = S_{ki} \quad \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}} = -\gamma_{ki}$$

which means that we can rewrite:

$$\frac{\partial P_k}{\partial P_{ki}} \frac{P_{ki}}{P_k} = \frac{S_{ki} \gamma_{ki}}{\sigma}$$

We can do the same on the Government side:

$$\frac{\partial P_{k,G}}{\partial P_{ki}} \frac{P_{ki}}{P_{k,G}} = \frac{\partial P_{k,G}}{\partial G_k} \frac{\partial G_k}{\partial Y_{ki}} \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{G_k}{P_{k,G}} \frac{P_{k,G}}{P_{ki}} \frac{P_{ki}}{Y_{ki}} = \frac{\partial P_{k,G}}{\partial G_k} \frac{G_k}{P_{k,G}} \frac{\partial G_k}{\partial Y_{ki}} \frac{Y_{ki}}{G_k} \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}}$$

where **(Proofs pending)**:

$$\frac{\partial P_{k,G}}{\partial G_k} \frac{G_k}{P_{k,G}} = -\frac{1}{\sigma} \quad \frac{\partial G_k}{\partial Y_{ki}} \frac{Y_{ki}}{G_k} = S_{ki}^G \quad \frac{\partial Y_{ki}}{\partial P_{ki}} \frac{P_{ki}}{Y_{ki}} = -\gamma_{ki}$$

and thus we can rewrite:

$$\frac{\partial P_{k,G}}{\partial P_{ki}} \frac{P_{ki}}{P_{k,G}} = \frac{S_{ki}^G \gamma_{ki}}{\sigma}$$

Substituting these expressions into the one of γ_{ki} :

$$\begin{aligned}
\gamma_{ki} &= \varepsilon - \frac{\varepsilon - \sigma}{Y_{ki}} \left(C_{ki} \frac{S_{ki} Y_{ki}}{\sigma} + G_{ki} \frac{S_{ki}^G Y_{ki}}{\sigma} \right) \Rightarrow \gamma_{ki} = \varepsilon - \frac{\varepsilon - \sigma}{\sigma} \gamma_{ki} \frac{C_{ki} S_{ki} + G_{ki} S_{ki}^G}{Y_{ki}} \Rightarrow \\
&\Rightarrow \gamma_{ki} \left(1 + \frac{(S_{ki} C_{ki} + S_{ki}^G G_{ki})(\varepsilon - \sigma)}{\sigma Y_{ki}} \right) = \varepsilon \Rightarrow \gamma_{ki} = \frac{\sigma \varepsilon Y_{ki}}{\sigma Y_{ki} + (S_{ki} C_{ki} + S_{ki}^G G_{ki})(\varepsilon - \sigma)} \Rightarrow \\
&\Rightarrow \gamma_{ki}^{-1} = \frac{1}{\varepsilon} + \frac{(\varepsilon - \sigma)}{\varepsilon \sigma} \frac{S_{ki} C_{ki} + S_{ki}^G G_{ki}}{Y_{ki}} \Rightarrow 1 - \gamma_{ki}^{-1} = \frac{\varepsilon - 1}{\varepsilon} + \frac{(\varepsilon - \sigma)(\varepsilon - 1)}{\sigma(\varepsilon - 1)\varepsilon} \frac{S_{ki} C_{ki} + S_{ki}^G G_{ki}}{Y_{ki}} \Rightarrow \\
&\Rightarrow \mu_{ki} = (1 - \gamma_{ki}^{-1})^{-1} = \frac{\varepsilon}{\varepsilon - 1} \left[1 - \frac{\frac{\varepsilon}{\sigma} - 1}{\varepsilon - 1} \frac{S_{ki} C_{ki} + S_{ki}^G G_{ki}}{Y_{ki}} \right]^{-1}
\end{aligned}$$

D Government Problem and First-Order Condition

The government chooses the allocation of procurement shares B_{ki} to maximize total consumption C . The household is subject to a budget constraint given by

$$P^C C = w - T$$

where T is the tax that the government collects. At the same time, government expenditures are equal to tax revenue:

$$T = P^G G$$

with G given. These two constraints allow us to rewrite consumption as

$$C = \frac{w - P^G G}{P^C}$$

Henceforth, the problem of the government becomes,

$$\max_{\{B_{ki}\}_{k,i}} \frac{w - P^G G}{P^C}$$

To find the optimal B_{ki} , we take the derivative of C with respect to B_{ki} and set it equal to zero.

$$\frac{\partial C}{\partial B_{ki}} = \frac{-\partial P^C / \partial B_{ki} (w - P^G G) + P^C G \cdot \partial P^G / \partial B_{ki}}{(P^C)^2} = 0$$

Multiplying through by P^C and substituting the expression for C , we obtain a condition relating the derivatives of P^C and P^G with respect to B_{ki}

$$\frac{\partial P^C}{\partial B_{ki}} C = -\frac{\partial P^G}{\partial B_{ki}} G.$$

Now, we rewrite the previous expression in terms of elasticities to obtain

$$\Rightarrow \frac{\partial P^C}{\partial B_{ki}} \frac{B_{ki}}{P^C} \left(\frac{\partial P^G}{\partial B_{ki}} \frac{B_{ki}}{P^G} \right)^{-1} = - \frac{P^G G}{P^C C}.$$

E Elasticity Representation

To ease notation and interpretation, we define the following elasticities:

$$\begin{aligned}\Phi_{ki}^s &= \frac{\partial P^s}{\partial B_{ki}} \cdot \frac{B_{ki}}{P^s} \quad (\text{aggregate elasticity}), \\ \varphi_{h,ki}^s &= \frac{\partial P_h^s}{\partial B_{ki}} \cdot \frac{B_{ki}}{P_h^s} \quad (\text{elasticity at sector level}), \quad s \in \{C, G\} \\ \varphi_{hj,ki} &= \frac{\partial s_{hj}}{\partial B_{ki}} \cdot \frac{B_{ki}}{P_{hj}} \quad (\text{elasticity at firm level})\end{aligned}$$

Using this notation, we can express the first order condition as,

$$(1) \quad \frac{\Phi_{ki}^C}{\Phi_{ki}^G} = - \frac{P^G G}{P^C C}$$

As we assume CES aggregation across sectors and firms, the aggregate elasticities and the elasticities at the sector level on the consumer side can be expressed as weighted averages of the elasticities at the firm level,

$$\Phi_{ki}^C = \sum_h S_h^C \varphi_{h,ki}^C, \quad \varphi_{h,ki}^C = \sum_j S_{hj}^s \varphi_{hj,ki}.$$

From the government side, we also account for an extra direct effect on the same sector.

$$\Phi_{ki}^G = \sum_h S_h^G \varphi_{h,ki}^G, \quad \varphi_{h,ki}^G = \sum_j S_{hj}^G \varphi_{hj,ki} + \mathbb{1}_{h=k} S_{ki}^G$$

These elasticities allow us to rewrite the first order condition of the problem of the government, equation (1), in terms of shares and elasticities at the firm level.

$$(2) \quad \frac{P^G G}{P^C C} = - \frac{\sum_h S_h^C \sum_j S_{hj}^C \varphi_{hj,ki}^C}{\sum_h S_h^G \sum_j S_{hj}^G \varphi_{hj,ki}^G + S_k^G S_{ki}^G}$$

F Elasticities proof

Defining $\Phi_{ki}^s = \frac{\partial P^s}{\partial B_{ki}} \frac{B_{ki}}{P^s}$, $\phi_{h,ki}^s = \frac{\partial P_h^s}{\partial B_{ki}} \frac{B_{ki}}{P_h^s}$, and $\phi_{hj,ki}^s = \frac{\partial P_{hj}^s}{\partial B_{ki}} \frac{B_{ki}}{P_{hj}^s}$, where $s = \{C, G\}$:

$$\begin{aligned} \frac{\partial P^C}{\partial B_{ki}} \frac{B_{ki}}{P^C} &= \frac{\partial \left(\sum_l A_l P_l^{1-\sigma} \right)^{\frac{1}{1-\sigma}}}{\partial B_{ki}} \frac{B_{ki}}{P^C} = \left(\sum_l A_l (P_l^C)^{1-\sigma} \right)^{\frac{\sigma}{1-\sigma}} \left(\sum_h A_h (P_h^C)^{-\sigma} \frac{\partial P_h^C}{\partial B_{ki}} \right) \Rightarrow \\ \Rightarrow \Phi_{ki}^C &= (P^C)^\sigma \left(\sum_h A_h (P_h^C)^{1-\sigma} \frac{\partial P_h^C}{\partial B_{ki}} \frac{1}{P_h^C} \right) \frac{B_{ki}}{(P^C)^\sigma} = (P^C)^{\sigma-1} \left(\sum_h A_h (P_h^C)^{1-\sigma} \frac{\partial P_h^C}{\partial B_{ki}} \frac{B_{ki}}{P_h^C} \right) \Rightarrow \\ \Rightarrow \Phi_{ki}^C &= \frac{\sum_h A_h (P_h^C)^{1-\sigma} \phi_{h,ki}}{\sum_l A_l (P_l^C)^{1-\sigma}} = \sum_h \frac{V_h^C \mu_h^{1-\sigma}}{\sum_l V_l^C \mu_l^{1-\sigma}} \phi_{h,ki} \Rightarrow \Phi_{ki}^C = \sum_h S_h^C \phi_{h,ki} \end{aligned}$$

Similarly,

$$\phi_{h,ki}^C = \sum_j S_{hj}^C \phi_{hj,ki} \quad \forall h$$

Regarding P^G :

$$\begin{aligned} \frac{\partial P^G}{\partial B_{ki}} \frac{B_{ki}}{P^G} &= \frac{B_{ki}}{P^G} \left(\sum_l B_l (P_l^G)^{1-\sigma} \right)^{\frac{\sigma}{1-\sigma}} \sum_h B_h (P_h^G)^{-\sigma} \frac{\partial P_h^G}{\partial B_{ki}} \Rightarrow \\ \Rightarrow \Phi_{ki}^G &= (P^G)^{\sigma-1} \sum_h B_h (P_h^G)^{1-\sigma} \frac{\partial P_h^G}{\partial B_{ki}} \frac{B_{ki}}{P_h^G} \Rightarrow \\ \Rightarrow \Phi_{ki}^G &= \frac{\sum_h B_h (P_h^G)^{1-\sigma} \phi_{h,ki}}{\sum_l B_l (P_l^G)^{1-\sigma}} = \sum_h \frac{V_h^G \mu_h}{\sum_l V_l^G \mu_l} \phi_{h,ki} \Rightarrow \Phi_{ki}^G = \sum_h S_h^G \phi_{h,ki} \end{aligned}$$

where

$$\frac{\partial P_k^G}{\partial B_{ki}} \frac{B_{ki}}{P_k^G} = \frac{\sum_j B_{kj} P_{kj}^{1-\epsilon} \phi_{kj,ki}}{\sum_m B_{km} P_{km}^{1-\epsilon}} + \frac{B_{ki} P_{ki}^{1-\epsilon}}{\sum_m B_{km} P_{km}^{1-\epsilon}} \Rightarrow \phi_{k,ki}^G = \sum_j S_{hj}^G \phi_{kj,ki} + S_{ki}^G \quad h = k$$

$$\phi_{h,ki}^G = \sum_j S_{hj}^G \phi_{hj,ki} \quad h \neq k$$

and thus, putting the results altogether:

$$\Phi_{ki}^C = \frac{\partial P^C}{\partial B_{ki}} \frac{B_{ki}}{P^C} = \sum_h S_h^C \phi_{h,ki} \Rightarrow \Phi_{ki}^C = \sum_h S_h^C \sum_j S_{hj}^C \phi_{hj,ki}$$

$$\Phi_{ki}^G = \frac{\partial P^G}{\partial B_{ki}} \frac{B_{ki}}{P^G} = \sum_h S_h^G \phi_{h,ki} \Rightarrow \Phi_{ki}^G = \sum_h S_h^G \sum_j S_{hj}^G \phi_{hj,ki} + S_k^G S_{ki}^G$$

and so, back to the FOC:

$$\frac{P^G G}{P^C C} = - \frac{\sum_h S_h^C \sum_j S_{hj}^C \phi_{hj,ki}}{\sum_h S_h^G \sum_j S_{hj}^G \phi_{hj,ki} + S_k^G S_{ki}^G}$$

G Demand rewritting

$$C_{ki} = A_{ki} A_{ki} (P_k^C)^{\varepsilon - \sigma} P_{ki}^{-\varepsilon} (P^C)^\sigma C \Rightarrow P_{ki} C_{ki} = A_{ki} A_{ki} (P_k^C)^{\varepsilon - 1 + 1 - \sigma} P_{ki}^{1 - \varepsilon} (P^C)^{\sigma - 1} P^C C \Rightarrow$$

$$\Rightarrow P_{ki} C_{ki} = A_k \left(\frac{P_k^C}{P^C} \right)^{1 - \sigma} A_{ki} \left(\frac{P_{ki}}{P^C} \right)^{1 - \varepsilon} P^C C \Rightarrow C_{ki} = P_{ki}^{-1} S_k^C S_{ki}^C P^C C$$

and similarly

$$G_{ki} = P_{ki}^{-1} S_k^G S_{ki}^G P^G G$$

And going back to the expression:

$$S_{ki}^C \left(1 + \frac{G_{ui}}{C_{ki}} \right)^{-1} + S_{ki}^G \left(1 + \frac{C_{ki}}{G_{ki}} \right)^{-1} \equiv R$$

We have:

$$R = S_{ki}^C \left(1 + \frac{S_k^G S_{ki}^G P^G G}{S_k^C S_{ki}^C P^C C} \right)^{-1} + S_{ki}^G \left(1 + \frac{S_k^C S_{ki}^C P^C C}{S_k^G S_{ki}^G P^G G} \right)^{-1} \Rightarrow$$